

HAMZA ALI

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PROFESSIONAL SUMMARY

Final-year Computer Science student specializing in Data Science at the University of Engineering & Technology (UET) Mardan, with all coursework, Final Year Project, and thesis requirements completed and currently awaiting final semester results. Passionate about data science, machine learning, and artificial intelligence, with practical experience gained through internships and academic projects. Skilled in Python, SQL, Power BI, PostgreSQL, Pandas, NumPy, and Scikit-learn, with hands-on experience in data analysis, predictive modelling, database systems, and business intelligence. Proven ability to work with real-world datasets, develop machine learning solutions, and create interactive dashboards that support data-driven decision-making. Eager to apply analytical and technical skills in Data Science, Machine Learning, and Analytics roles while continuing to expand expertise in AI and advanced data technologies.

EXPERIENCE

June 2024-September 2024 **DATA SCIENCE INTERN**

INFINITYBITS, ISLAMABAD, Pakistan

Assisted in end-to-end dashboard development and data preprocessing initiatives

- Collaborated on Power BI dashboard creation, designing KPI tracking dashboards that streamlined business reporting and reduced report generation time.
- Performed advanced data preprocessing and visualization tasks on structured datasets, preparing data for machine learning model development and statistical analysis.
- Generated actionable business insights from structured datasets, supporting data-driven decision-making processes across departments.

August 2024 **DATA SCIENCE INTERN**

PRODIGY INFOTECH (ONLINE)

Developed machine learning solutions and data analysis workflows for real-world datasets

- Performed comprehensive exploratory data analysis (EDA) and data cleaning on real-world datasets, improving data quality by 35% and identifying key patterns for downstream analysis.
- Built and optimized decision tree classifier for customer purchase prediction, achieving 82% accuracy and enabling targeted marketing strategies.
- Conducted comprehensive sentiment analysis on social media data using NLP techniques, processing 10,000+ records to extract business insights and customer sentiment trends.
- Created interactive visualizations using Matplotlib and Power BI, transforming complex data patterns into actionable business insights for stakeholders.

EDUCATION

2022-2026 **UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN**

Mardan, KPK, Pakistan

Bachelor's in computer science (Specialization in Data Science); Major in Data Science

- Awards: Merit Based Scholarship (only 2 awarded to class)
- Awards: Selected In Dean's Award List, 2025-2026 (Appreciation Certificate) (Only 5 awarded to class)
- Achievement: SmogNet Datathon Competition (Got 1ST Position)
- Achievement: Final Year Project selected in DGST (Directorate of Science & Technology) for research funding
- Assistant: Collaborated With Teacher to Help In their Routines, Lectures and Labs

CGPA: 3.33 / 4.00

PROJECTS AND RESEARCH

CARBON FOOTPRINT ESTIMATOR SYSTEM (FINAL YEAR PROJECT)

- Comprehensive environmental impact analytics and emission estimation system
- Developed a scalable system to estimate daily carbon emissions and environmental impact, integrating multiple data sources and calculation methodologies.
- Designed comprehensive relational database structure using PostgreSQL to store emission data, user profiles, and environmental metrics with optimal indexing.
- Implemented complex emission calculation logic handling multiple sources (transportation, energy consumption, waste) with real-time data processing capabilities.
- Focused on environmental impact analytics, providing users with personalized carbon footprint reports and actionable recommendations for sustainability.
- Integrated data visualization dashboards to track emission trends over time and compare against sustainability benchmarks.

BANKING & FINANCIAL INTELLIGENCE SYSTEM

- Designed comprehensive relational database architecture using PostgreSQL with ACID transaction modeling and deadlock handling for mission-critical banking operations.
- Built advanced fraud detection model using Credit Card Fraud dataset and machine learning algorithms, achieving 94% accuracy in identifying fraudulent transactions.
- Created interactive Power BI dashboards to monitor KPIs and fraud trends in real-time, enabling proactive risk management and fraud prevention strategies.
- Integrated machine learning predictions into reporting pipeline, automating fraud alerts and enabling dynamic risk assessment.

BLOOD DONOR PREDICTION & RE-DONATION SYSTEM

- Built Logistic Regression model to predict donor re-donation probability, leveraging historical donation patterns and demographic data for optimized donor outreach.
- Optimized model performance using AUC score evaluation and threshold tuning, achieving 87% sensitivity in identifying high-probability re-donors.
- Applied automated model selection techniques (TPOT) to identify optimal algorithms and feature combinations, reducing manual tuning time by 40%.
- Performed comprehensive feature engineering and data preprocessing, creating meaningful derived features that improved model interpretability and performance.

BANK MARKETING CLASSIFICATION MODEL

- Developed decision tree classification model to predict customer subscription to bank products, enabling targeted marketing campaigns and improved campaign ROI.
- Performed comprehensive exploratory data analysis (EDA) and preprocessing to identify key customer segments and behavioral patterns.
- Improved model accuracy from 75% to 88% using advanced feature selection techniques, including recursive feature elimination and correlation analysis.

TECHNICAL SKILLS

- Programming & Databases: Python (Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow), SQL, PostgreSQL, C/C++
- Data Analytics & BI: Exploratory Data Analysis (EDA), Data Cleaning & Preprocessing, Statistical Analysis, Power BI (DAX, KPI Dashboards, Data Modeling)
- Machine Learning: Logistic Regression, Decision Trees, Random Forest, Feature Engineering, Model Evaluation, TPOT
- Data Engineering: ETL Pipelines, Data Pipeline Development, Database Design, Data Warehousing
- Specialized Tools: Jupyter Notebook, Git, Tableau, Power BI, Microsoft Fabric

CERTIFICATIONS

- [IBM Introduction to Data Science Specialization](#)
- **IBM Data Science Specialization** (Completed 7 Courses): [What is Data Science?](#) , [Data Analysis with Python](#), [Databases and SQL for Data Science with Python](#), [Python for Data Science](#), [Data Science Methodology](#), [Tools for Data Science](#), [Python Project for Data Science](#)
- [Google Crash Course on Python](#)
- [Microsoft Data Analysis & Visualization with Power BI](#)
- [Python Data Structures](#)
- [Programming for Everybody \(Getting Started with Python\)](#)
- [Ai For Everyone](#)
- [Foundations: Data, Data, Everywhere](#)

OTHER

- **Languages:** English (Professional), Urdu (Native).
- **Soft Skills:** Problem-Solving, Data Storytelling, Business Acumen, Team Collaboration, Communication
- **Key Strengths:** End-to-end ML project implementation, Business intelligence dashboard development, Statistical analysis, Data-driven decision making